MIA5150 / GAP ANALYSIS

MIA5150 gaps and experimental results

A visual evidence review | 24 October 2026

The main gap was a like-for-like final-selection comparison, not missing retrieval work. The local study now measures it; original hosted comparisons, human labels and independent evaluation remain open. [S1, S2, S6]

- Several absence claims were outdated: full-public-query retrieval, the original four catalog sizes and three-seed sensitivity already existed. [S1]

- New primary evidence: 200 queries, five methods and four sizes produce 4,000 planned outcomes. These are repeated conditions on 200 queries, not 4,000 independent tasks. [S2, S4]

- At 6,006 skills, Hybrid exactly selected 11/200 gold sets versus 9/200 for BM25. The weighted difference is +1.02 percentage points, with a nominal 95% paired interval from -2.50 to +4.98. No clear winner follows. [S2, S3]

Scope and evidence boundary

Measured local scope: all five discovery methods and sizes 10, 100, 1,000 and 6,006, with one Qwen3 0.6B Q8 selector. This is constrained metadata selection, not skill execution. [S2, S4, S8]

The original 80 method-size-model cells remain in a versioned crosswalk. Local substitutes do not complete unrun hosted cells; submitted and revised scopes remain separate. [S1, S7]

Panel role

Exposed public-test sample; exploratory evidence

All 4,392 public SkillRet test queries had already been exposed. This frozen subset is not a fresh holdout. [S1, S8]

Decision

Retain the bounded local findings and failures. Next priorities are genuine human labels, independent requests and credential-enabled hosted comparisons. [S6]

Separate full-public retrieval check

n = 4,392 exposed queries; N = 6,006 skills. Three catalog seeds agree; results shown once.

Method	Recall@5	MRR@5
BM25	0.477345	0.523201
Vector	0.517304	0.587739
Hybrid	0.587014	0.654994

Retrieval only, not final selection. This panel overlaps the 200-query panel; counts are not pooled. [S11]

Audit baseline c4f4889fb3ec6864898d3ba07d08c30c438e3e12

Evidence approved 2026-10-24T18:28:34Z

Gap coverage by scope

Status is scoped to the evidence below. Already present means the baseline allegation was too broad; closed means a bounded local requirement is supported, not that the whole original proposal is complete. [S1, S6]

Requirement and scope	Status	Evidence and remaining gap
Original sizes and full-query retrieval Scope: both	Already present	Earlier studies already covered 10/100/1,000/6,006 and all 4,392 public queries. [S1] Remaining: Do not present their absence as a newly fixed gap.
Matched final selection Scope: both	Closed	A common 200-query, five-method, four-size local grid replaces reliance on two-query diagnostics. [S1, S2] Remaining: Closed only for this 0.6B deployment and exposed panel.
Fresh public holdout claim Scope: both	Outdated claim	The old 4,192 reserved-ID premise no longer applies after full public-test studies. [S1, S8] Remaining: Independently sourced, untouched requests are still needed.
Identity and scoring controls Scope: both	Closed	The new harness uses stable IDs, strict selected sets and explicit failure denominators. [S4, S6] Remaining: This does not retroactively repair historical runs.
Efficiency and reproducibility Scope: both	Partial	Tokens, preflight limits and corrected fresh timing are recorded with provenance. [S2, S5] Remaining: Timing is valid-cohort evidence; unknown failed usage stays unknown.
Robustness and generalization Scope: both	Partial	Separate 24-query two-model comparison completed; Qwen3 0.6B also covers three seeds. [S1, S2, S10] Remaining: These exposed sensitivity panels do not establish independent generalization.
Human scenarios and task execution Scope: both	Open	Generated silver labels and gold-count strata are not human semantic labels. [S6, S8] Remaining: Human adjudication, cascade ordering and actual execution remain pending.
Original hosted model matrix Scope: original	Not tested	Local Qwen results are explicitly substitutes, not completed original model cells. [S6, S7] Remaining: Hosted credentials are absent; zero paid calls were made.

Already present identifies baseline coverage, not a new fix. Outdated claim identifies an allegation contradicted by the reviewed baseline. Closed records an evidenced closure; partial, open, deferred, not tested and out of revised scope remain distinct qualitative outcomes.

Method and comparison design

The controlled unit is a query evaluated under each method and size. Catalogs retain its known positive skills, so scaling measures conditional distractor difficulty rather than natural catalog coverage. [S4]

1. **Freeze the comparison**
   Pin queries, qrels, nested catalogs, models and protocol. [S4]

2. **Discover candidates**
   Full listing, BM25, vector, single-domain vector or hybrid. [S4]

3. **Select an explicit skill set**
   The same local selector returns constrained JSON IDs; no tools execute. [S4, S8]

4. **Score and retain every outcome**
   Separate retrieval, final exact match, preflight failure, invalid output and timing. [S2, S5]

Comparability and sample

Same 200 queries, seed 42, metadata view, library size and selector. Failures remain in all-planned quality denominators; full listing has no ranked retrieval Recall@5. [S2, S4]

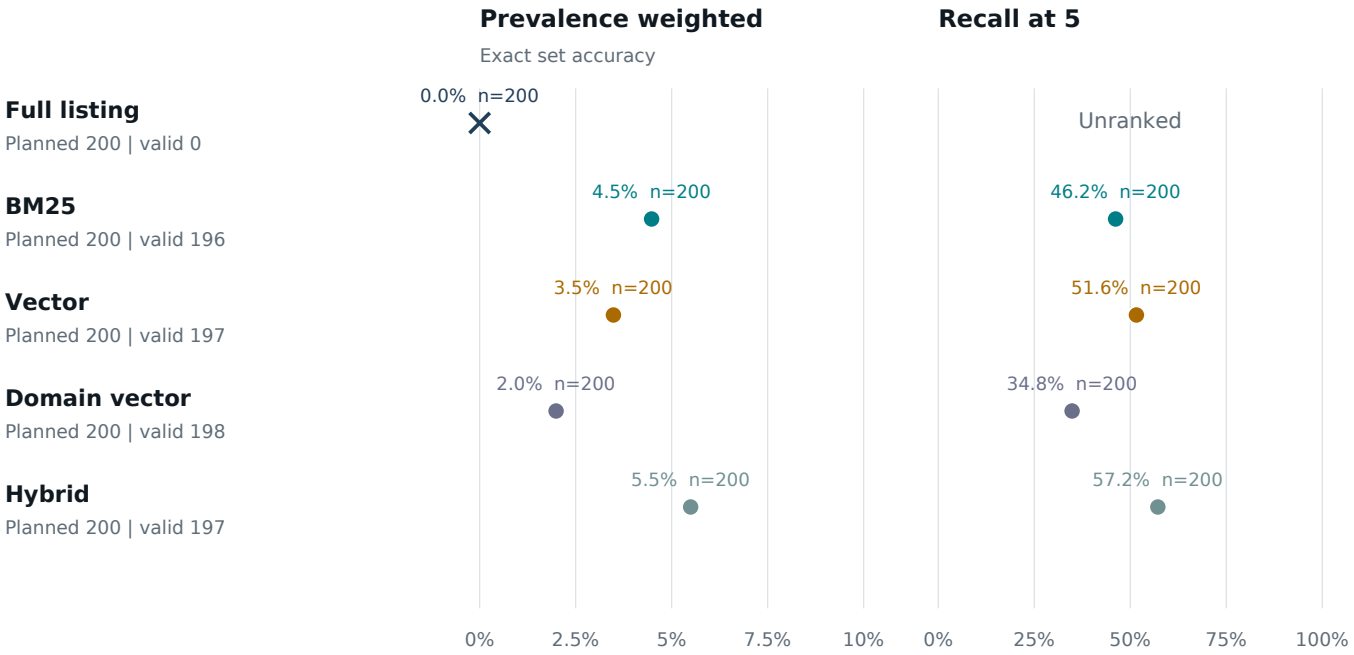
Panel role	Exposed public-test sample; exploratory evidence
Panel queries	200
Scenario composition	98 single-positive, 78 two-positive and 24 three-positive queries; seed 42
Exposure	All 4,392 public SkillRet test queries had already been exposed. This frozen subset is not a fresh holdout. [S1, S8]

- Final exact-set accuracy is prevalence-weighted across gold-count strata. Retrieval Recall@5 and MRR@5 are sample-macro means; they answer different questions. [S2, S3]

- The single-domain arm cannot retain all gold skills when they span domains. This is a structural limitation, not just a router error. [S9]

Matched retrieval and selection results

Full catalog of 6,006 skills



X = preflight exclusion; all-planned score remains zero

Figure 1. Aligned outcomes from the selected matched comparison. Accuracy basis: prevalence-weighted exact-set accuracy. For the weighted estimate, n is the unweighted query sample size, not a binomial denominator. Independent zero-based panel scales: accuracy 0-10%; recall 0-100%. No confidence intervals were supplied; none are inferred. [S2]

Weighting: Official gold-count proportions 2143/4392, 1703/4392 and 546/4392; sample strata 98/78/24. The table retains raw exact counts and valid-output denominators.

Method	Exact sets	Valid outputs	Other outcomes	MRR at 5
Full listing	0/200	0/200	200 preflight	Unranked
BM25	9/200	196/200	4 invalid	0.532
Vector	7/200	197/200	3 invalid	0.582
Domain vector	4/200	198/200	1 other unsuccessful; 1 invalid	0.399
Hybrid	11/200	197/200	3 invalid	0.637

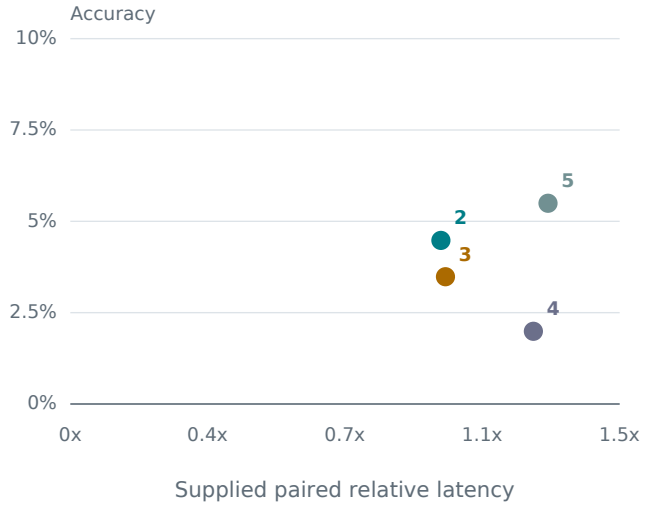
Exact sets and valid outputs use planned-query denominators. Other unsuccessful stages may include routing or guard failures; they are not all model-inference crashes. Preflight exclusion is not a reasoning test.

- Higher retrieval coverage did not guarantee a correct final set. At full scale, raw exact successes are BM25 9, Vector 7, Domain vector 4 and Hybrid 11 out of 200. [S2]

- All 200 full-listing cases were excluded by the configured 8,192-token preflight at 1,000 and 6,006 skills. Their zero all-planned score is an operational result, not a test of the model reasoning over those prompts. [S2, S4]

Latency and token tradeoffs

Paired latency versus accuracy



Logical tokens versus accuracy

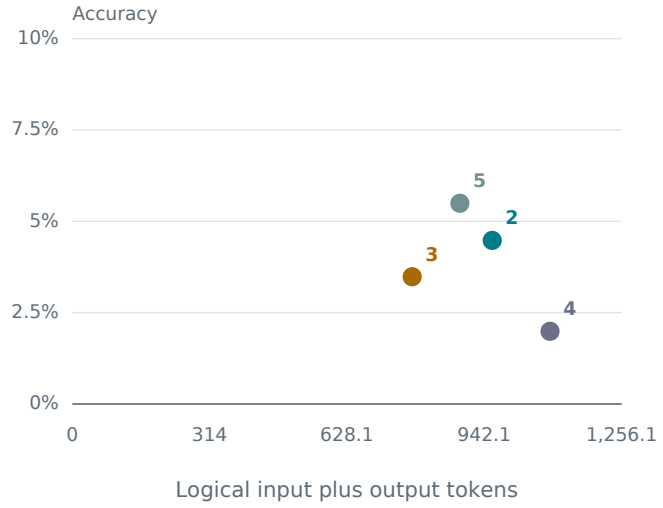


Figure 2. Supplied ratios use method-specific paired-valid cohorts, not ratios of unconditional means. The table distinguishes paired n from the absolute timing n.

Figure 3. Logical input/output estimates are not physical campaign charges. Embedding tokens are separate; routing is not added again. Missing costs are omitted. Markers identify table rows.

Accuracy: prevalence-weighted exact-set accuracy, on independent 0-10% axes.

Accounting basis: Valid-output logical input/output means include fresh routing where applicable. They are not an invoice or total physical campaign usage; embedding tokens are separate.

Timing scope: Supplied ratios use paired, valid, fresh cases and a matched implementation. Absolute means use their own fresh-valid n. Preflight, failure, setup and replay times are excluded. Recorded timing values were not independently remeasured.

Method	Fresh valid ms	Paired relative	Logical input	Logical output	Embedding tokens
1. Full listing	Not supplied n=0	Not measured	Not supplied	Not supplied	0
2. BM25	704.1 n=147	1x paired n=147	910.7	49.4	0
3. Vector	708.5 n=148	1.01x paired n=146	726.5	50.6	268.8
4. Domain vector	878.9 n=149	1.25x paired n=147	1,033.7	58.6	267.3
5. Hybrid	907 n=148	1.29x paired n=146	834.1	52.1	268.8

Paired reference BM25 with the same selector and library size; Paired valid, fresh cases from the retained implementation; no route reuse. Pair counts differ by method.

Cost and paired-cohort evidence [S2, S5].

- Full listing at 6,006 skills required a mean 411,291 locally tokenized prompt tokens. Those preflight tokens were not consumed by the model and do not appear as a paid token bill. [S2]

- Latency comparisons cover smaller paired valid cohorts, not all 200 planned cases. Treat small differences as deployment measurements, not hardware-independent speed claims. [S2, S5]

Interpretation and closure

What the evidence supports

- Candidate coverage, final set selection and operational feasibility are separate bottlenecks. A larger candidate-recall score alone cannot establish better end-to-end skill selection. [S2]

- At 6,006 skills, Hybrid minus BM25 is +1.02 percentage points; its nominal 95% paired interval [-2.50, +4.98] includes zero. [S3]

Uncertainty and threats to validity

- Intervals use 2,000 query-paired bootstrap draws within gold-count strata, reused across sizes and contrasts. Per-cell intervals are not supplied; no error bars are invented. Family-eight tail intervals are nominal, not a guarantee of power. [S3]

- All queries are exposed, qrels may omit plausible alternatives, and gold-conditioned catalogs differ from natural coverage. Repeated methods and sizes do not increase independent n. [S1, S4, S8]

- Under this study's category mapping, 69% of queries have all positives in one domain. The oracle single-domain macro-recall ceiling is 84.75%, before routing and ranking errors. [S9]

Separate model sensitivity

n = 24 balanced exposed queries at 6,006 skills, seed 42. Not pooled with the primary panel.

Method	Qwen3 0.6B Q8 exact sets	Qwen3.5 4B Q4 exact sets
Full listing	Preflight excluded	Preflight excluded
BM25	2/24	4/24
Vector	0/24	1/24
Domain vector	0/24	0/24
Hybrid	1/24	1/24

Architectures, training and quantization differ; this is not a causal parameter-count experiment. [S10, S2]

Revised scope closure

The local primary comparison is measured. Methodological controls and the scope crosswalk close specific reporting gaps; they do not make the overall research claim complete. [S2, S6]

Original scope disclosure

Uncompleted original hosted cells are blocked by missing credentials (zero paid calls). The 4B model has only 24-query sensitivity, not a 200-query primary. Fresh independent requests, human semantic overlap/cascade labels and actual task execution remain unvalidated. [S6, S7, S8]

Remaining actions

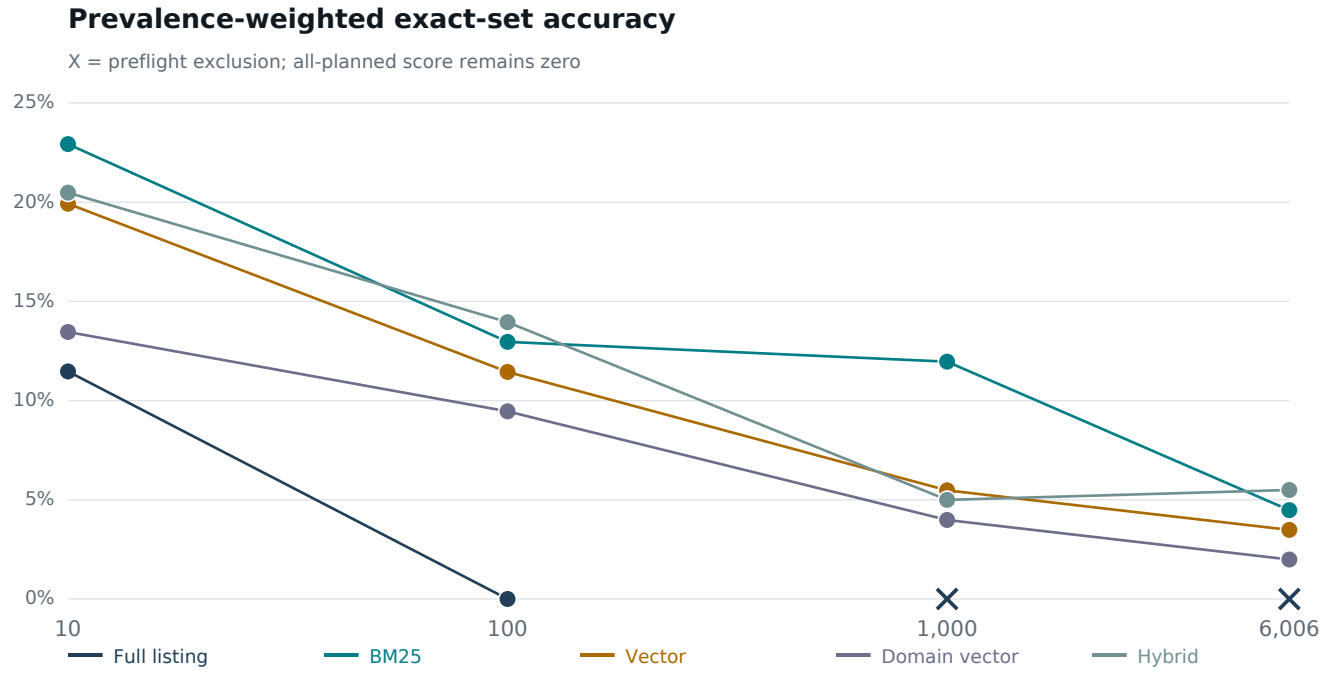
- Review sensitivity limits: retain separate query counts, seed coverage and model-configuration caveats. [S2, S10]

- Obtain genuine human authorship/adjudication and an independently sourced panel; keep generated silver probes labeled. [S6]

- Team-review the paper and presentation research drafts before course submission. [S6, S8]

Supplementary evidence

Selection accuracy across library sizes



The same 200 queries recur at all four sizes. Points show all-planned, prevalence-weighted exact-set accuracy; library size is log-scaled. Listing zeros at 1,000 and 6,006 are preflight exclusions. [S2, S4]

Method	10 skills	100 skills	1000 skills	6006 skills
Full listing	11.5% (178/200)	0.0% (196/200)	0.0% (0/200)	0.0% (0/200)
BM25	22.9% (196/200)	13.0% (195/200)	12.0% (197/200)	4.5% (196/200)
Vector	19.9% (193/200)	11.4% (198/200)	5.5% (194/200)	3.5% (197/200)
Domain vector	13.5% (181/200)	9.5% (197/200)	4.0% (196/200)	2.0% (198/200)
Hybrid	20.5% (197/200)	14.0% (198/200)	5.0% (192/200)	5.5% (197/200)

Cells show accuracy (valid/planned outputs). The same queries recur across sizes; counts are not pooled. Missing estimates break lines. The 0-25% scaling axis is independent of the main panels.

No confidence intervals were supplied for this scaling figure.

- At 100 skills, listing produced 196 valid outputs but zero exact gold sets. Context fit alone did not solve selection. [S2]

Sources and reproducibility

S1 Independent baseline gap audit | e8690e35

Open source - Baseline c4f4889f

S2 Matched selection and efficiency table | e8690e35

Open source - primary200_seed42; qwen3_06_q8

S3 Paired primary contrasts and uncertainty | e8690e35

Open source - 200 queries; 2,000 stratified paired resamples

S4 Frozen benchmark protocol | e8690e35

Open source - Primary panel, methods, sizes and scoring

S5 Timing and missing usage corrections | e8690e35

Open source - Valid cohorts, routed search and unknown usage

S6 Gap closure matrix | 0f4e9818

Open source - Scope-specific evidence and remaining requirements

S7 Original proposal crosswalk | e8690e35

Open source - All 80 historical cells, including unrun hosted cells

S8 Study reproduction and limits | 0f4e9818

Open source - Pinned assets, runtime and exposed-query boundary

S9 Domain routing diagnostics | e8690e35

Open source - Primary n=200 at each size

S10 Separate model and seed sensitivity | e8690e35

Open source - n=24; Qwen3 seeds 42/43/44; Qwen3.5 seed 42

S11 Separate full-public retrieval | 0f4e9818

Open source - 4,392 exposed queries; full catalog reported once across three seeds

S12 Independent raw audits | 0f4e9818

Open source - Published primary and local-sensitivity audits; full-retrieval receipt is separate

- Primary/sensitivity tables are pinned to e8690e35; full-public retrieval is pinned to 0f4e9818. Rebuild tables from archived ledgers with aggregate.py; rendering this PDF reruns no inference. [S1, S8, S11]

- The paper and presentation remain research drafts for team review. This visual report does not establish activation or task-execution success. [S6, S8]

- The published audit folder covers primary and local sensitivity. Full-public retrieval passed a separate independent receipt; it is not represented as part of that folder. [S11, S12]